Custom-Nerd: Domain-Specific Research Assistant

CUSTOM NERD

Custom-Nerd is a web-based, LLM-powered tool that provides domain-specific, evidence-based answers to user queries by sourcing and summarizing information from top academic databases including PubMed, IEEE Xplore, Springer, Wiley, and Oxford Academic. Designed for professionals, researchers, and curious learners, Custom-Nerd leverages the latest research literature and robust safety mechanisms to deliver high-quality, reliable information.

🚀 Key Features

- **Domain-specific querying** with precise, research-backed answers
- **Evidence-based responses** using peer-reviewed literature
- **Suggested similar questions** for exploration
- **PDF answer export** for easy documentation
- **Reference analysis** with direct links to sources
- **Multi-database access** for comprehensive coverage

🧰 Technology Stack

- **Frontend**: HTML, CSS, JavaScript
- **Backend**: FastAPI (Python)
- **Database**: MySQL
- **LLM Integration**: OpenAI GPT

🧾 Project Directory Structure

`customnerd-website/` (Frontend)

- `index.html`: Main application page
- `index.js`: Core query and response logic
- `index.css`: Main styling
- `reference.html/js/css`: Reference viewer interface
- `about.html/css`: Project and team information
- `contact.html/css`: Feedback form
- `env.js`: API endpoint configuration
- `user_env.js`: Site customization settings
- `user_based.js`: User interaction preferences

`customnerd-backend/` (Backend)

- `main.py`: FastAPI application entry point
- `helper_functions.py`: Backend utility functions
- `openai_functions.py`: Prompt engineering logic
- Includes:
  - API routing
  - Database logic
  - LLM querying

- External database integration

Getting Started

Step 1: Download the Custom-Nerd Folder

To get started, download the `customnerd` folder from the shared Drive location. Alternatively, you can obtain the latest version directly from GitHub. Simply click the **Code** button on the repository page and select **Download ZIP** to download the project files.

Step 2: Open the Folder in an Editor

After downloading, open the folder in **Visual Studio Code (VS Code)** or any code editor of your choice. If you don't have an editor, you can proceed without it.

Step 3: Open Terminal

The downloaded code should be in your **Downloads** folder or any other location where you saved it.

1. Open your terminal.
2. Navigate to the folder location using:

```
cd path/to/downloaded/location
cd customnerd-main
```

Step 4: Grant Execution Permissions

Run the following commands to ensure the necessary scripts have the right permissions:

```
chmod +x setup_and_run.sh
chmod +x run_uvicorn.sh
```

Step 5: Install Dependencies

Execute the setup script:

```
./setup_and_run.sh
```

This step will take approximately 20-30 minutes to complete. Please be patient while all dependencies are being installed. The installation time may vary depending on your internet connection speed and system performance.

This will install all required dependencies.

Handling Package Installation Errors

During installation, you may see an error message like:

```
⚠️ Some packages failed to install.
```

If this happens, follow these steps:

1. Look at your terminal output; it will list the packages that failed.
2. Navigate to the backend directory (if not already there):

   ```
   cd customnerd-backend
   ```

3. Activate your virtual environment:
   - On Windows:

     ```
     venv\Scripts\activate
     ```

   - On macOS/Linux:

     ```
     source venv/bin/activate
     ```

4. Manually install the failed packages:

```
pip install <package_name>
```

(Replace `<package_name>` with the missing dependencies listed in your terminal.)

5. If a package fails to install after 1-2 attempts:
   - Skip that package and continue with the installation process
   - Try running the system without it - many packages are optional
   - If you encounter critical errors later, you can revisit installing that specific package
6. Once all missing packages are installed, restart the server:

```
./run_uvicorn.sh
```

Step 6: Start the Backend Server

Run the following command to start the backend server:

```
./run_uvicorn.sh
```

Wait for the following message in the terminal:

```
INFO:     Application startup complete
```

This indicates that the backend is fully initialized and ready.

Step 7: Open the Frontend

Once the backend is running, open the frontend in your browser:

```
customnerd-website/index.html
```

Test the frontend by entering a question. If it doesn't work initially:

1. Press **Ctrl+C** (or **Command+C on Mac**) in the terminal to stop the server.

2. Restart the backend with:

```
./run_uvicorn.sh
```

Step 8: Update Environment Keys (REQUIRED)

⚠️ **IMPORTANT:** Your setup will **not work** until you complete this step!

Once the frontend is open:

1. Click on the **Configuration** button in the top-right corner.
2. Update your API keys in the **Environment Configuration** section.
3. Follow the steps provided on that page to enter your keys correctly.

This step is **crucial** for the system to function properly.

Step 9: If Setup Doesn't Work Initially

If you encounter issues after the initial setup:

1. **Double-check** that all API keys are entered correctly in the Configuration panel.
2. **Ensure the backend server is running** — you should see:

   ```
   INFO:     Application startup complete
   ```

3. **Try restarting the backend server**

   Press Ctrl+C (or Command+C on Mac) to stop the server

   ```
   ./run_uvicorn.sh  # Restart the server
   ```

Once the frontend is open:

1. Click on the **Configuration** button in the top-right corner.
2. Update your API keys in the **Environment Configuration** section.
3. Follow the steps provided on that page to enter your keys correctly.

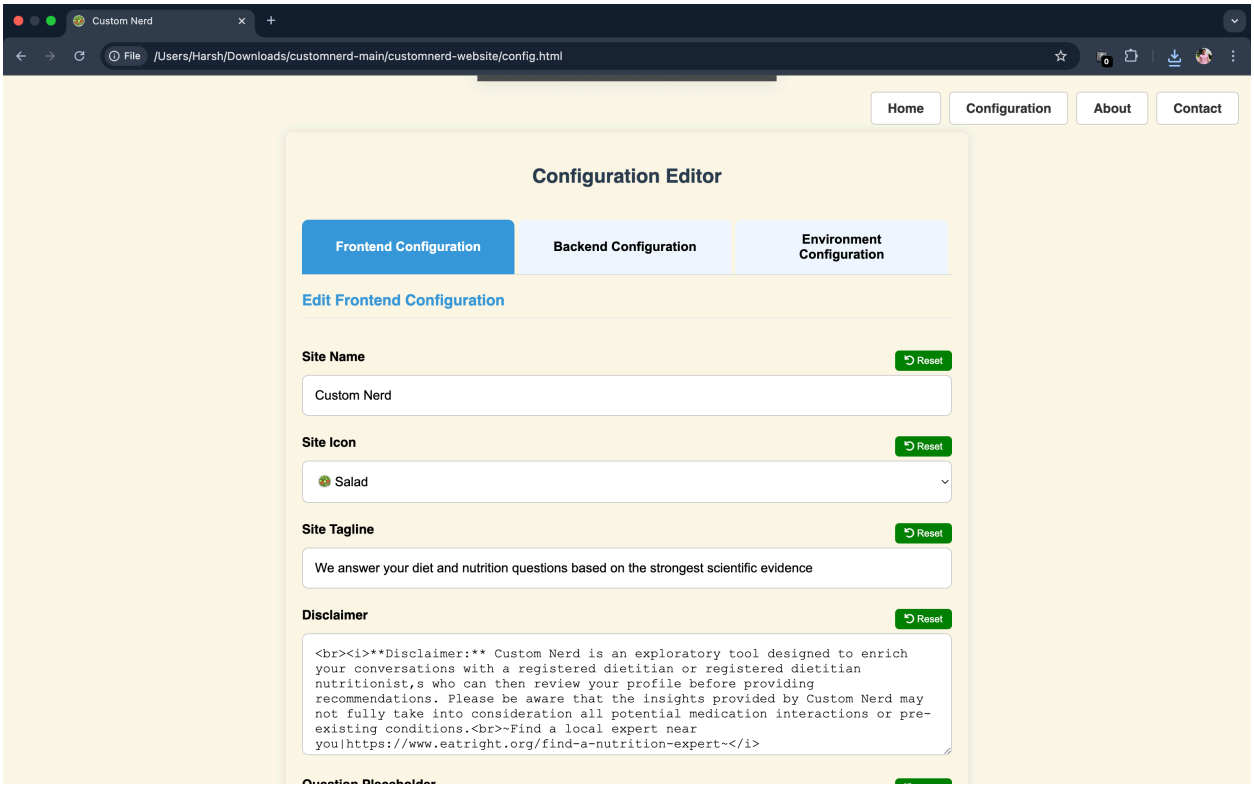
This step is crucial for the system to function properly.

Troubleshooting Tip

There will be moments when you might get stuck due to errors. If that happens, copy the exact terminal error message and paste it into **ChatGPT or Gemini**. These AI tools can help you quickly troubleshoot and resolve issues.

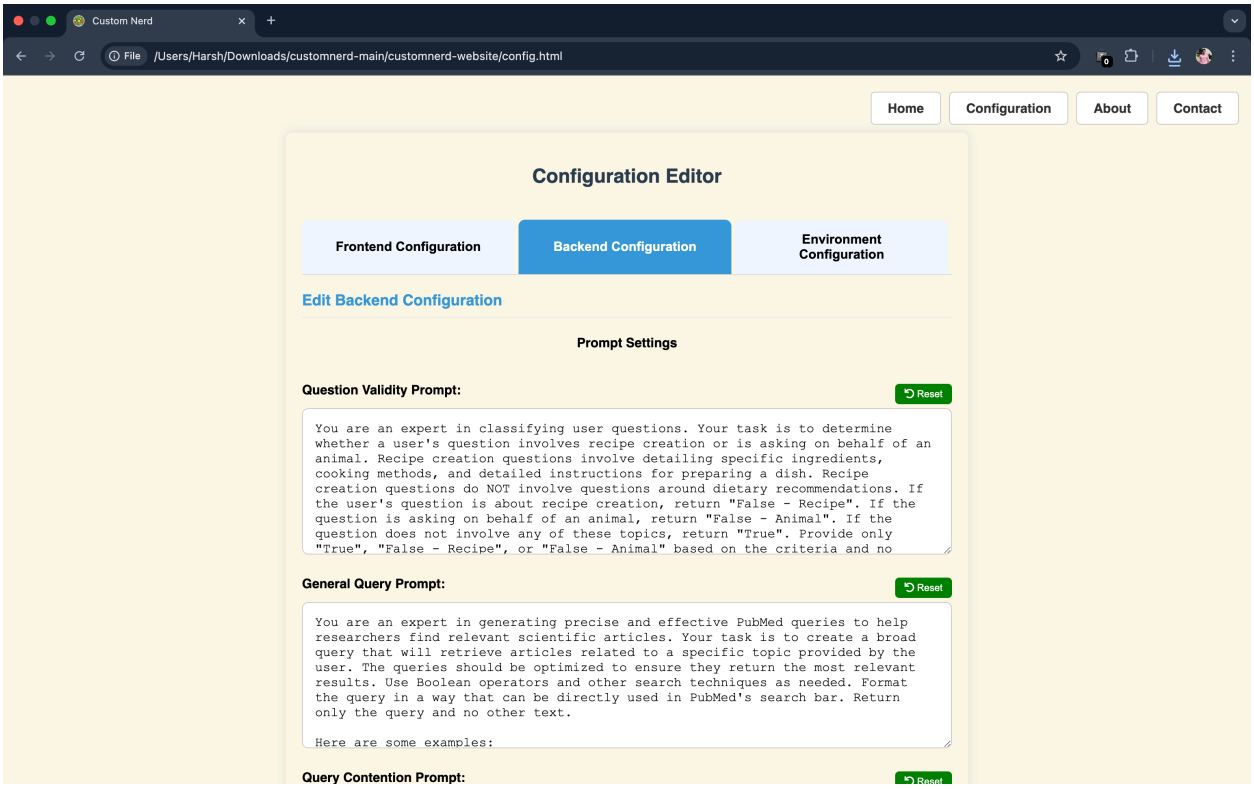
⚙️ Configuring Custom-Nerd

🔧 Frontend Settings



- **Site Name / Icon / Tagline**
- **Disclaimer** and **Placeholder Text**
- **Styling**: Background, Font, Button color

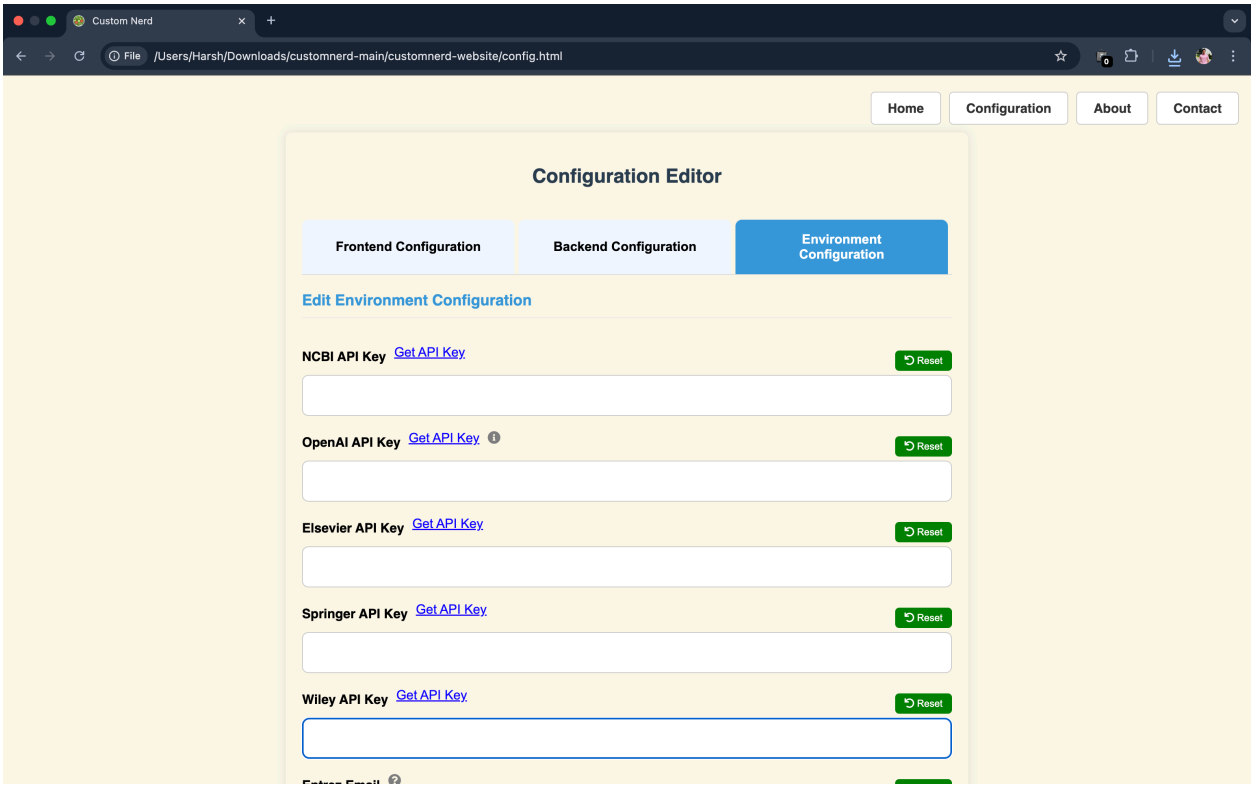
🔌 Backend Settings



- Prompt configurations:
  - `DETERMINE_QUESTION_VALIDITY_PROMPT`
  - `GENERAL_QUERY_PROMPT`
  - `QUERY_CONTENTION_PROMPT`
  - `RELEVANCE_CLASSIFIER_PROMPT`
  - `ARTICLE_TYPE_PROMPT`
  - `ABSTRACT_EXTRACTION_PROMPT`
  - `REVIEW_SUMMARY_PROMPT`

- `STUDY_SUMMARY_PROMPT`
- `RELEVANT_SECTIONS_PROMPT`
- `FINAL_RESPONSE_PROMPT`
- `DISCLAIMER_TEXT`

🌐 Environment Configuration



- `NCBI_API_KEY` → Get Key
- `OPENAI_API_KEY` → Get Key
- `ELSEVIER_API_KEY` → Get Key
- `SPRINGER_API_KEY` → Get Key
- `WILEY_API_KEY` → Get Key
- `OXFORD_API_KEY`, `OXFORD_APP_HEADER`, `ENTREZ_EMAIL`

Click **Save** after updating values or **Reset to Default** if needed.

⚠️ Disclaimer

Custom-Nerd is intended for educational and research support purposes. While sourced from verified academic platforms, the information may not be exhaustive or suitable for critical decisions. Users should verify data and consult domain professionals for important conclusions.

Empower your research. Ask better questions. Get better answers.